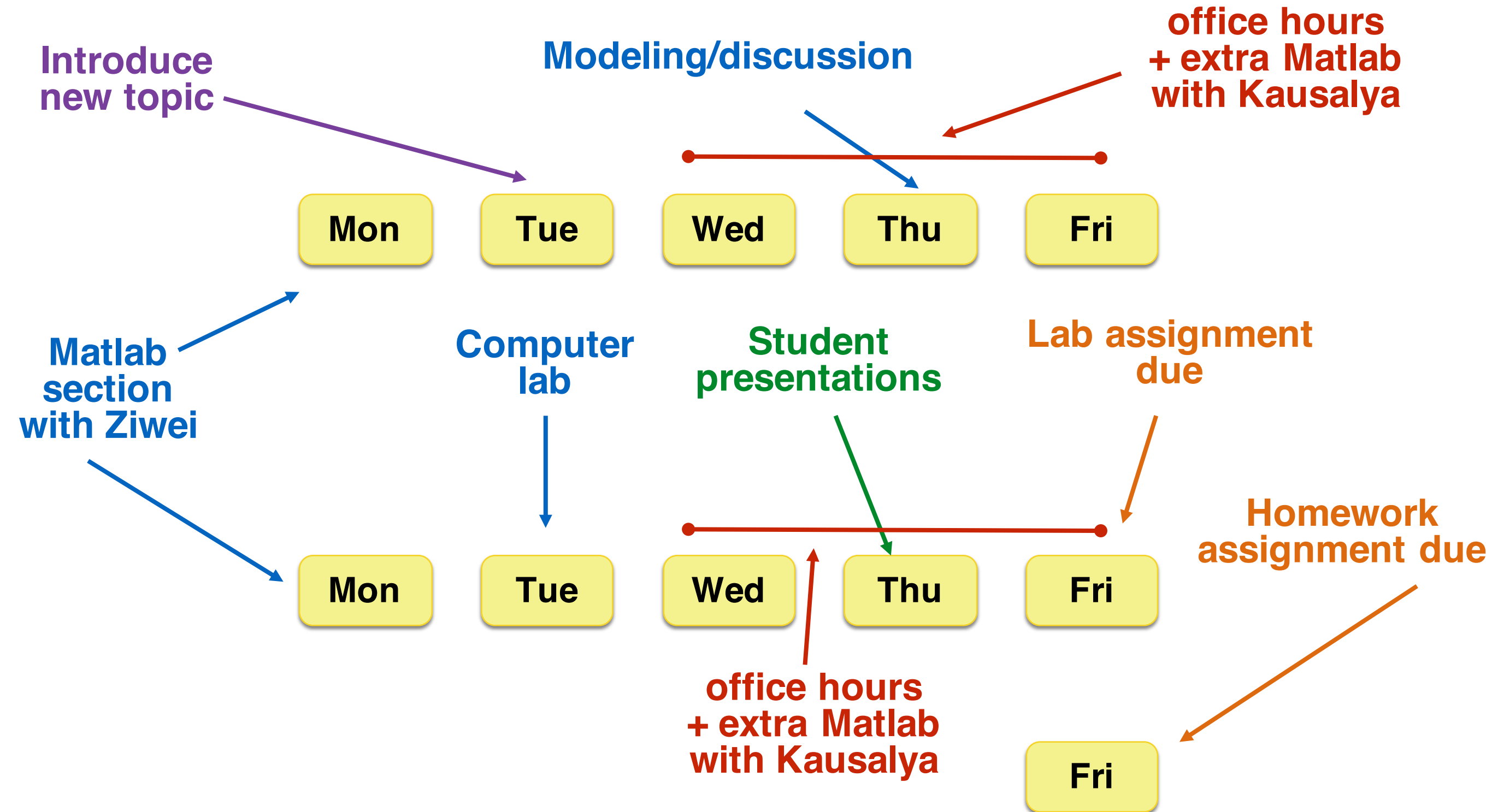


Lecture 3: overview

- **Admin about student presentations**
- **Continuation of Matlab basics**
- **Introduce flu pandemics**
- **The SIR model of Kermack and McKendrick (1927) – modeling activity!**
- **Google trends**

General two-week topic schedule



Admin: student presentations

- **Five lecture times (Feb 9, Feb 23, Mar 9, Mar 30, Apr 13)**
- **Sign-up done using a Google Doc. Link posted this evening. 3 - 4 slots per session. Must be done with in a team of three students.**
- **Think about finding a team, which of the five topics you'd like to present, and when works for you.**
- **Practice presentation and review slides with Kausalya, a new TF!**
- **Database of papers going up soon. Papers for next week will be online tonight. You must choose a different paper from other presenters.**

Course layout

Jan 24	Introduction	Mar 14	Spring Break
Jan 31	Topic 1: Epidemiology	Mar 21	Topic 4: Information theory
Feb 7		Mar 28	
Feb 14	Topic 2: Imaging	Apr 4	Topic 5: Game theory
Feb 21		Apr 11	
Feb 28	Topic 3: Random walks	Apr 18	Topic 6: Chaos or Projects
Mar 7		Apr 25	

Matlab loops & functions

Example script: summing.m

Example function: sumfunction.m

Topic 1: Epidemiology

Flu pandemics

Swine Flu (H1N1) Vaccine

13 January 2011 Last updated at 11:14 ET



By THE NEW YORK TIMES

Sixty more deaths, but flu 'may be starting to plateau'

Overview

A vaccine for the H1N1 strain of [swine flu](#) became available in October 2009 as the country girded for a second wave of the pandemic that broke out across the country and much of the world in the spring. Despite shortages and confusion in the initial weeks of the vaccination effort, by December enough doses were available that officials expanded the effort beyond the high-risk groups initially targeted: pregnant women, people under 24 years old or caring for infants, people with high-risk medical conditions and health-care workers.

TABLE OF CONTENTS

- [Overview](#)
- [Flu Returns, Vaccine Arrives](#)
- [Delivering the Vaccine](#)
- [The Vaccine's Impact](#)
- [An Anxious Public](#)
- [Meeting Worldwide Demand](#)
- [Frequently Asked Questions](#)

By the end of the year, over 85 million doses were available. By February 2010, there was so much unused swine flu vaccine in the world that rich nations, including the United States, were trying to get rid of their surpluses. But logistical difficulties meant that the world's poorest countries — a few still facing the brunt of the pandemic — were receiving very little of it.

Medical experts looking back on the course of the epidemic -- although it may be far from over -- have credited luck and a series of rapid but conservative decisions by federal officials with limiting the virus's toll. In December, federal health officials estimated that 10,000 Americans had died since April of swine flu, far below the range of 30,000 to 90,000 predicted by a presidential panel.

Sixty-two people died with flu in the UK in the past week, bringing the total to 112 since October, figures reveal.

However, the Health Protection Agency (HPA) said the rise could be due to delayed reporting, with many of the deaths occurring before Christmas.

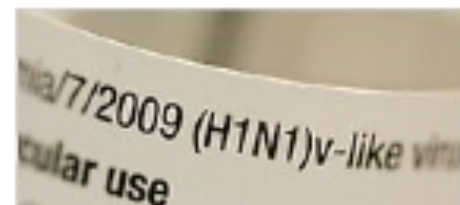
Most of those who died had swine flu. The majority were aged between 15 and 64, but nine children under 14 died.

An HPA spokesman said the number of people seeing their GP for flu-like illness was beginning to plateau.

Of the 81 deaths where data is available, 69 people were in risk groups for flu.

And of the 47 for whom vaccine status is known, 40 had not had this season's flu jab.

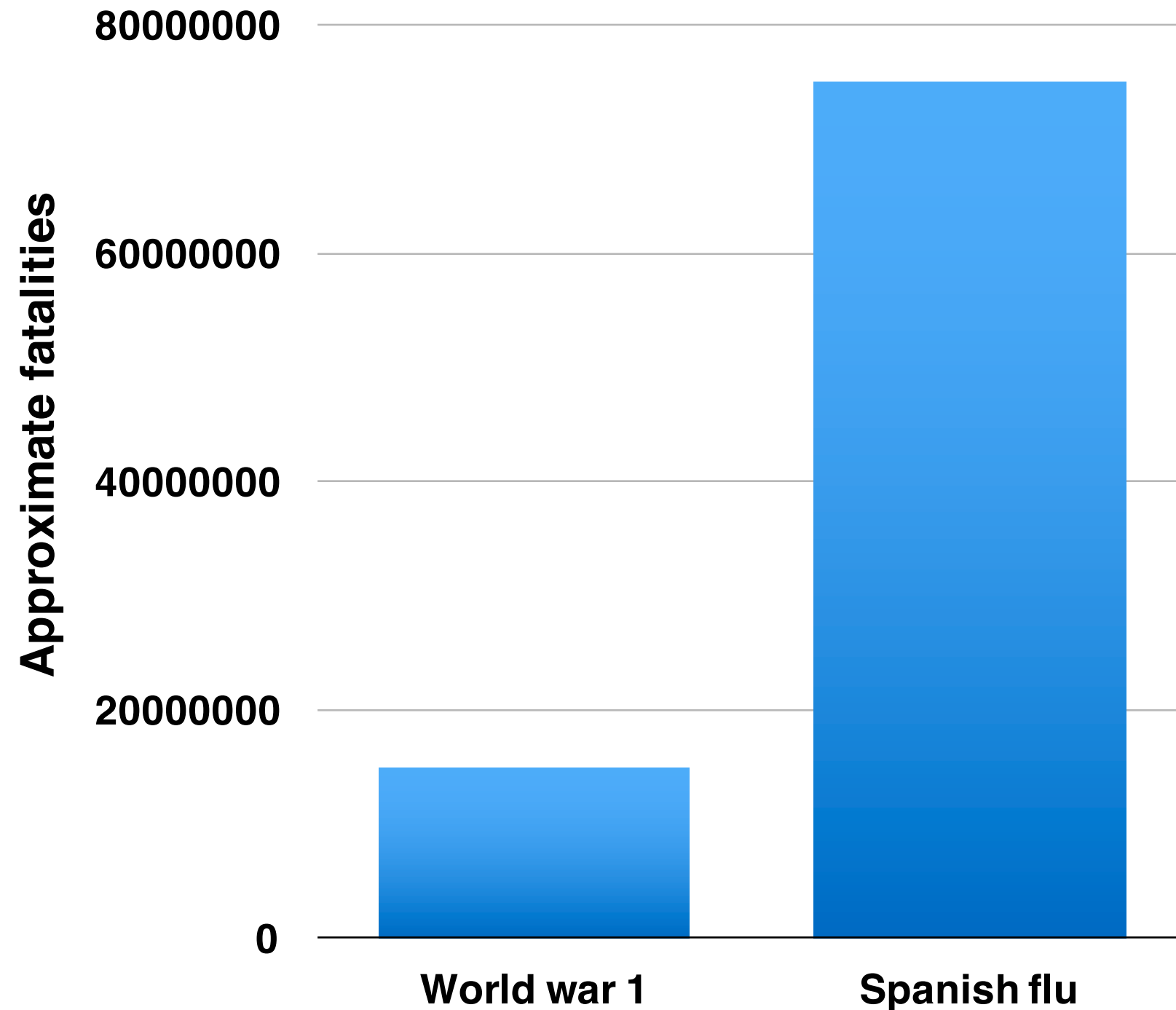
~ 14,000 deaths worldwide



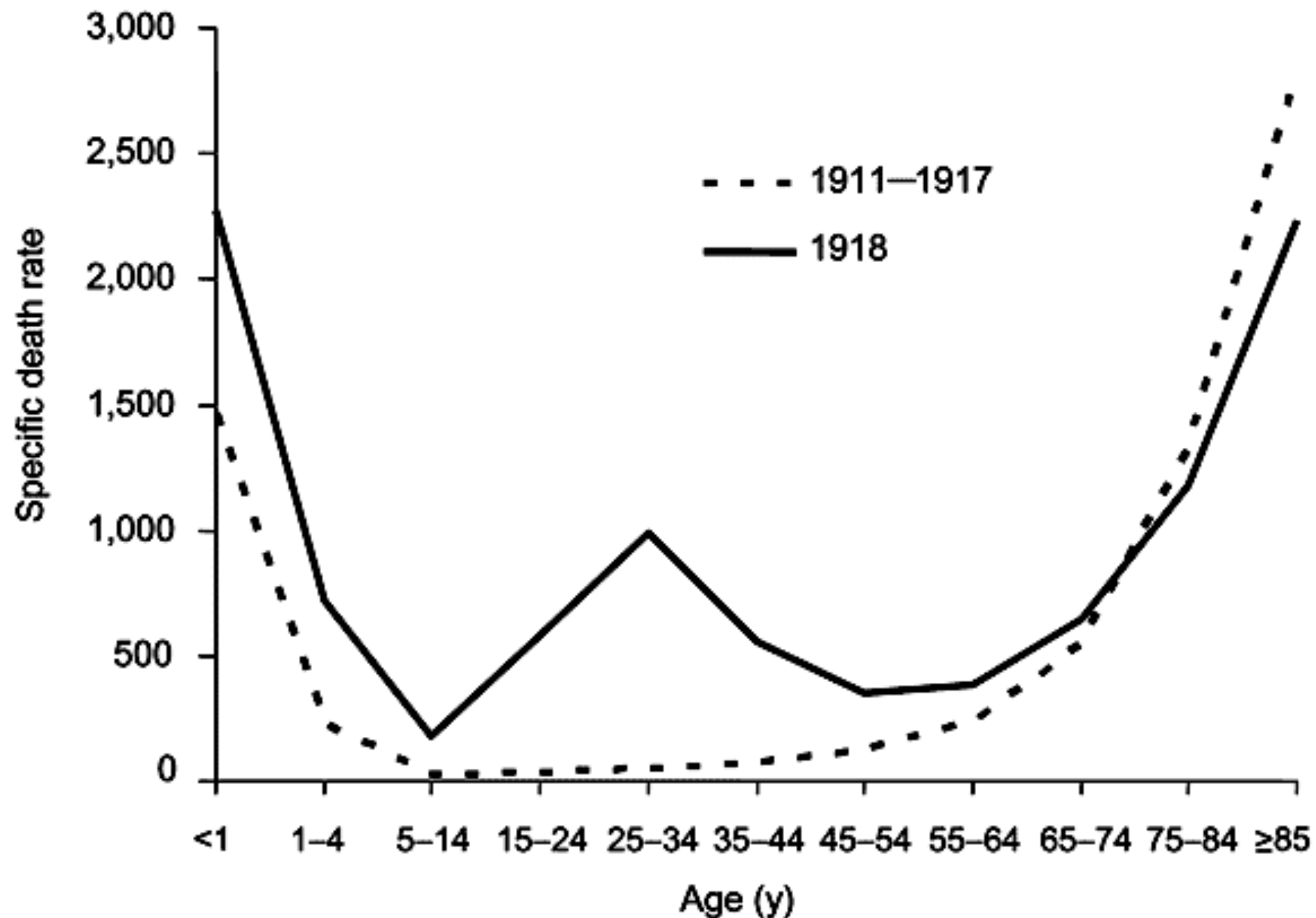
1918 “Spanish flu” pandemic

- **One of the worst flu outbreaks in history**
- **500 million infected across entire world**
- **50–100 million killed (3%–5% of world’s population)**
- **Unclear geographical origin, but likely accelerated by massive troop movement at the end of World War 1**
- **Free press in neutral Spain gave impression that Spain was hardest hit**

Significantly higher death toll than WW1

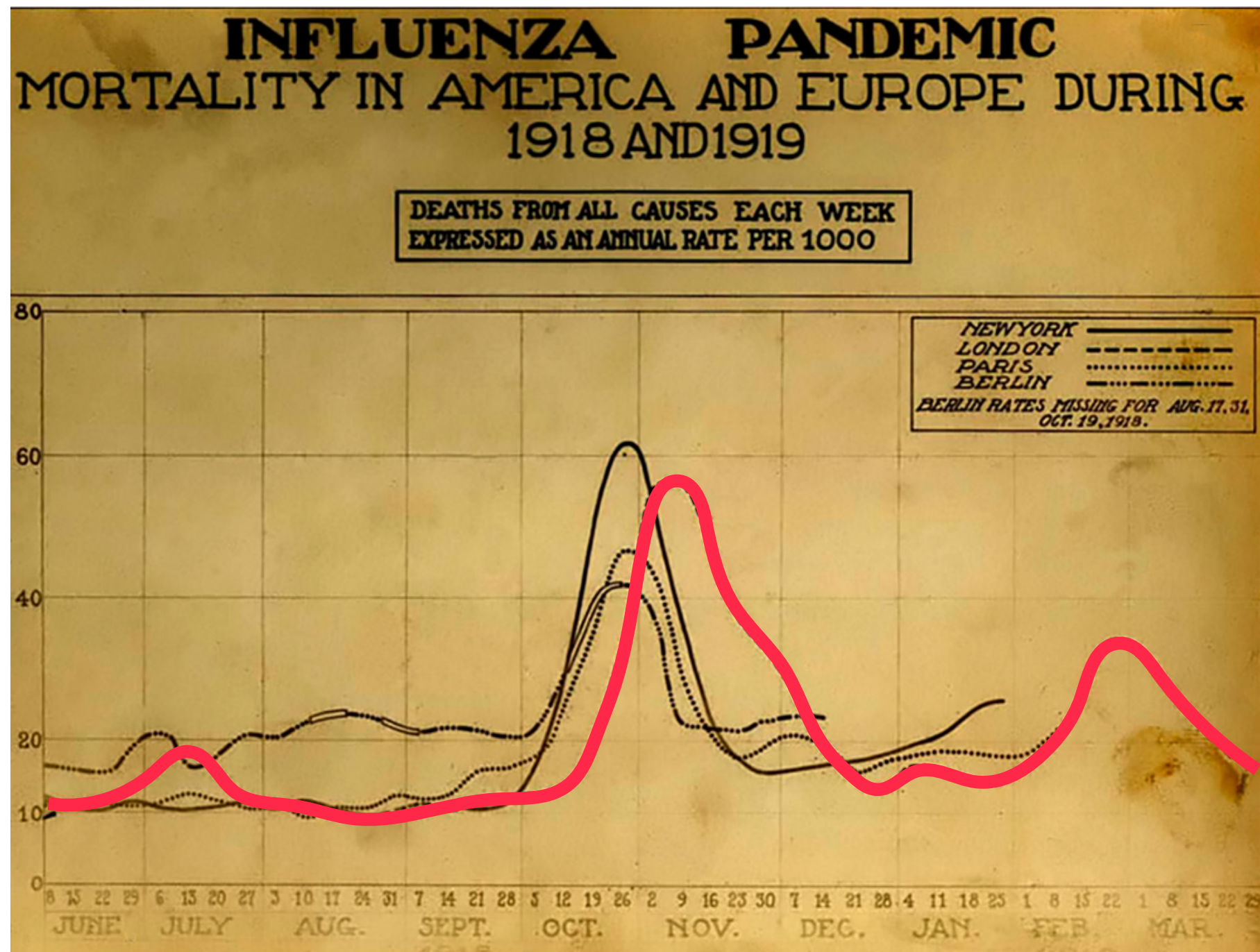


Mortality by age group

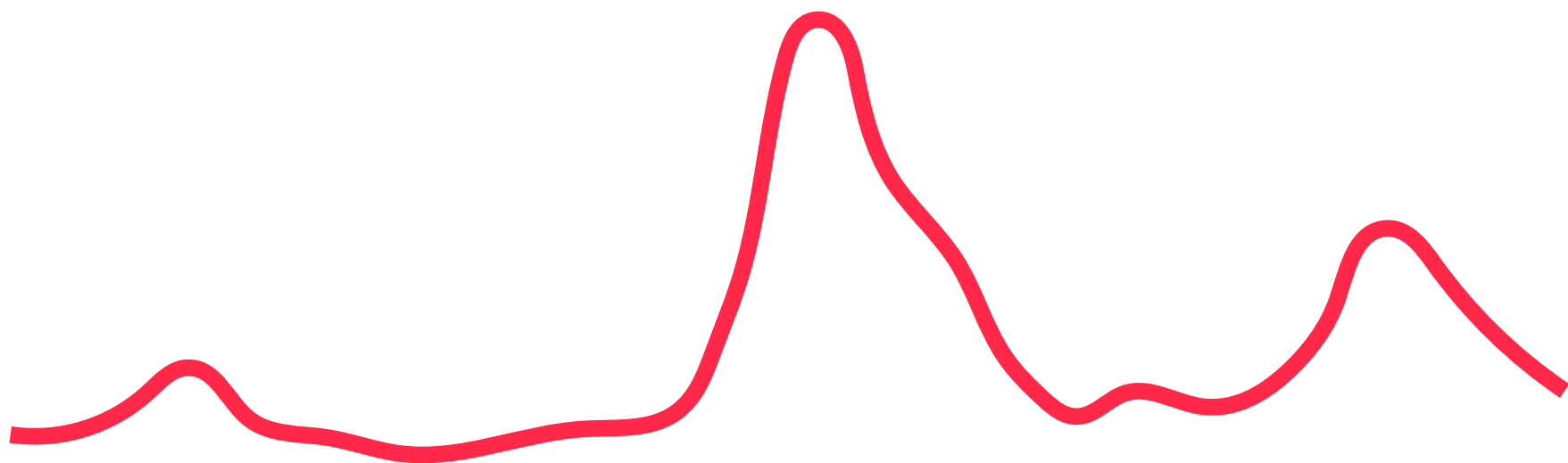


- Large number of deaths in healthy adults, due to over-response of the immune system

Mortality in different cities: similar patterns



Mortality in different cities: similar patterns



Just look at data for London: multiple similar peaks

Questions

- 1. Why are some flus pandemics (widespread epidemics) and some not? Why are pandemics so sudden?**
- 2. Why do flus end? Perhaps all people who can get it either recover or are “removed” from the population?**
- 3. How can we anticipate a flu pandemic in the early stages?**
- 4. What are the most effective vaccination strategies?**
- 5. Suppose someone in this room had bird flu? How worried should you be that you’ll be infected?**

A Contribution to the Mathematical Theory of Epidemics.

By W. O. KERMAK and A. G. MCKENDRICK.

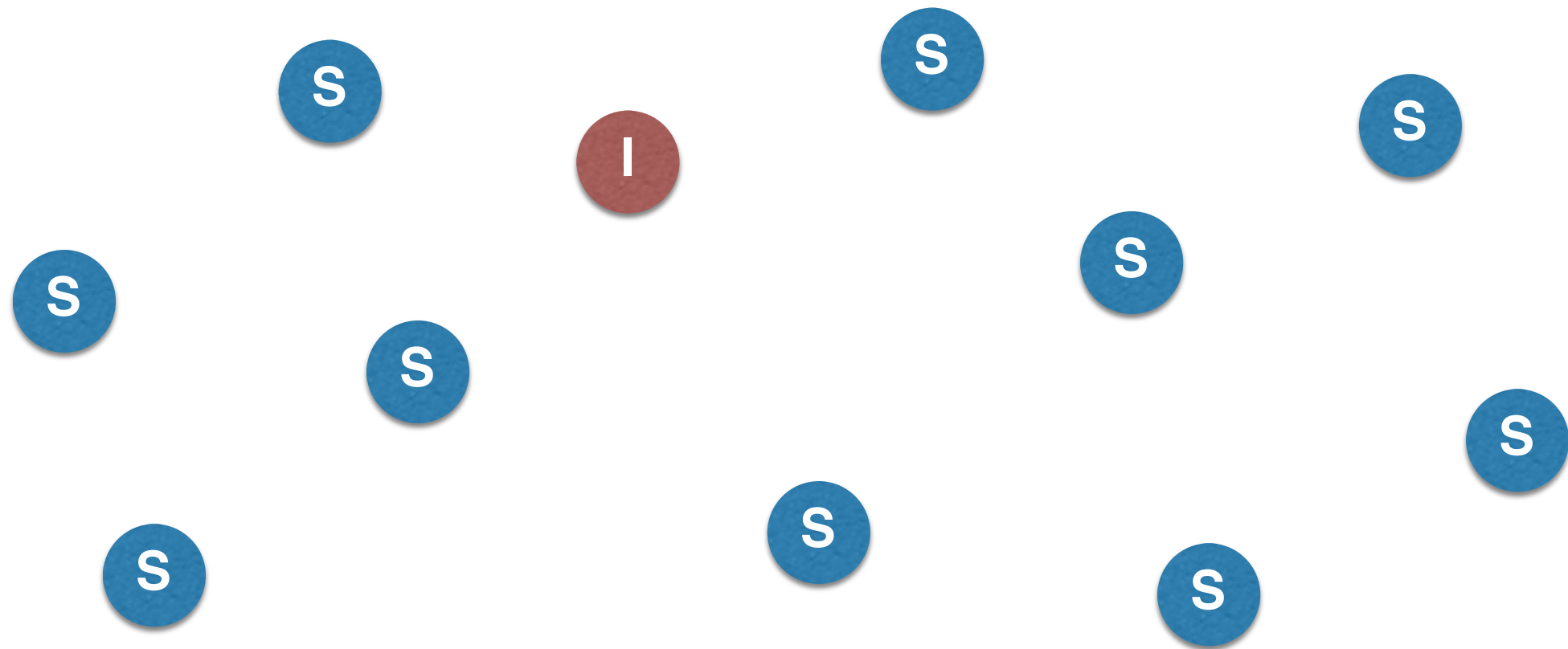
(Communicated by Sir Gilbert Walker, F.R.S.—Received May 13, 1927.)

(From the Laboratory of the Royal College of Physicians, Edinburgh.)

Introduction.

(1) One of the most striking features in the study of epidemics is the difficulty of finding a causal factor which appears to be adequate to account for the magnitude of the frequent epidemics of disease which visit almost every population. It was with a view to obtaining more insight regarding the effects of the various factors which govern the spread of contagious epidemics that the present investigation was undertaken.

General idea

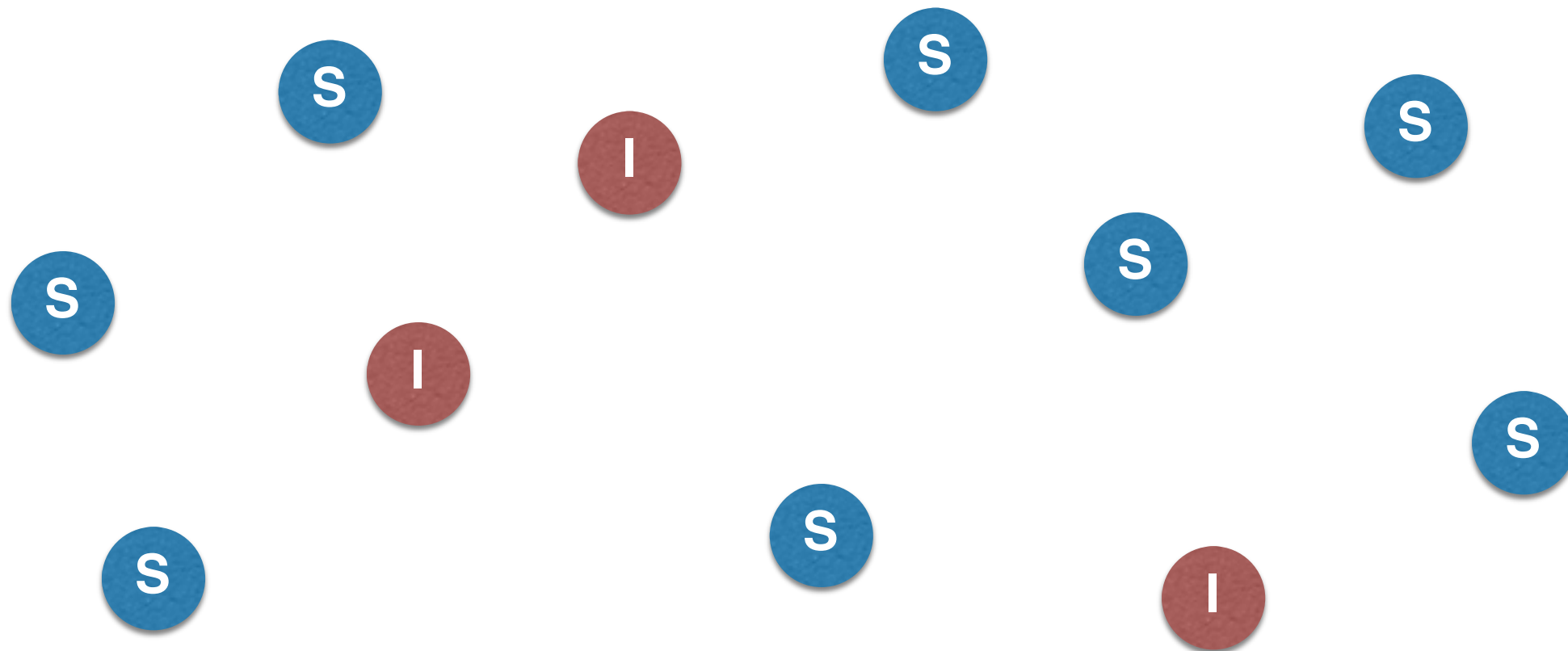


Three populations:

- **Susceptible** $S(t)$
- **Infected** $I(t)$
- **Removed** $R(t)$



General idea

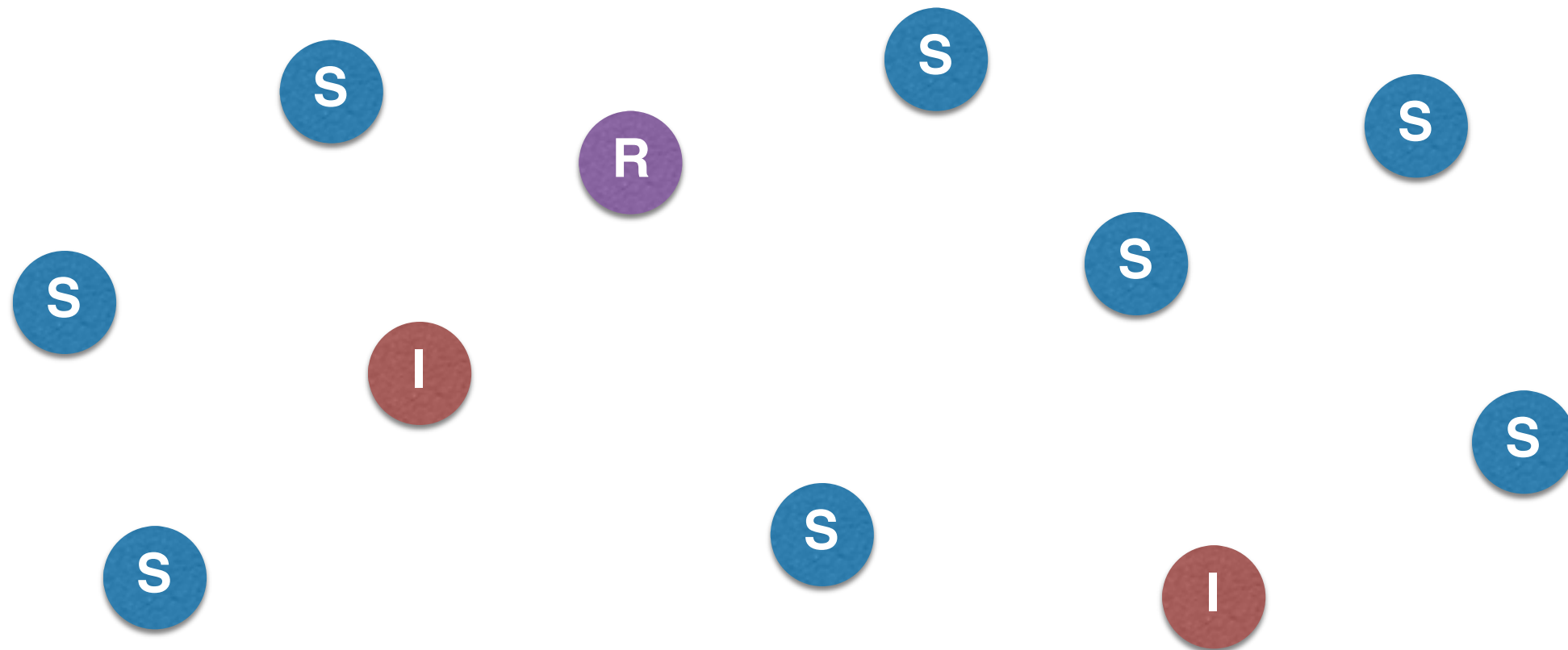


Three populations:

- **Susceptible** $S(t)$
- **Infected** $I(t)$
- **Removed** $R(t)$



General idea



Three populations:

- **Susceptible** $S(t)$
- **Infected** $I(t)$
- **Removed** $R(t)$



How do we turn this into equations?

x_t : Number of people not yet infected

y_t : Number of people infected

z_t : Number of people removed (recovered/died)

- Number of people infected at time t is

$$v_t = x_t - x_{t+1}$$

- Total people sick at time t is

$$y_t = \sum_{\theta=0}^t v_{t,\theta} = v_{t,0} + v_{t,1} + \dots + v_{t,t}$$

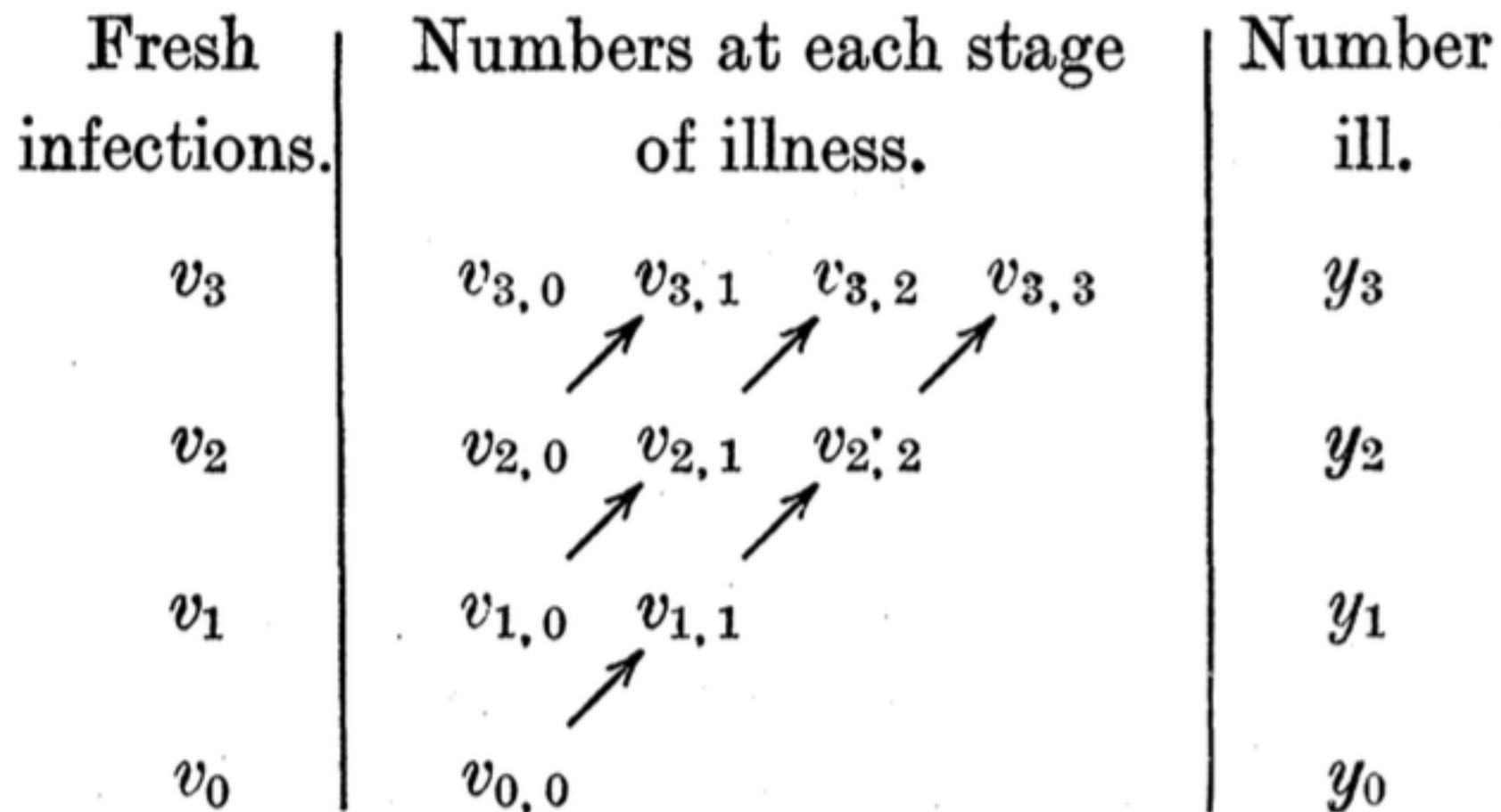
number of people sick at time t
who were infected at time θ

- Total population is

$$x_t + y_t + z_t = N$$

How do we turn this into equations?

The whole process is indicated in the following schema :—



$v_{t,\theta}$: number of people sick at time t
who were infected at time θ

y_t : total number of people sick at time t

v_t : number of people who get sick at time t

Calculus: assume that intervals become very small

$$x_t + y_t + z_t = N.$$

$$v_t = -\frac{dx_t}{dt},$$

$$\frac{dx_t}{dt} = -x_t \left[\int_0^t A_\theta v_{t-\theta} d\theta + A_t y_0 \right],$$

$$\frac{dz_t}{dt} = \int_0^t C_\theta v_{t-\theta} d\theta + C_t y_0,$$

$$y_t = \int_0^t B_\theta v_{t-\theta} d\theta + B_t y_0,$$

A_t is related to rates of infection and removal.

B_t and C_t are related to rate of removal.

Difficult solution

- The authors then attempt to solve these equations
- Before digital computers: couldn't use simulations to gain insight into solutions

- They manage to derive a formula for the size of an epidemic

$$-\log \frac{1-p}{1-\frac{y_0}{N}} = ANp$$

- Here, p is the proportion infected during an epidemic
- A bit too complicated for clear insight—is there any way to simplify the model?

Simpler model: constant rates

B. *Constant Rates.*

(10) Much insight can be obtained as to the process by which epidemics in limited populations run their peculiar courses, and end in final extinction, from the consideration of the special case in which ϕ and ψ are constants κ and l respectively.

In this case the equations are

$$\frac{dx}{dt} = -\kappa xy$$

$$\frac{dy}{dt} = \kappa xy - ly$$

$$\frac{dz}{dt} = ly$$

and as before $x + y + z = N$.

SIR Model (using modern notation)

$S(t)$: **Susceptible**

$I(t)$: **Infected**

$R(t)$: **Removed/recovered**

$$\frac{dS}{dt} = -\beta SI$$

Infection rate



$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

Recovery/death rate



**Let's develop this model
ourselves!**